

Code No: 125DT**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B.Tech III Year I Semester Examinations, May/June - 2019****COMPUTER NETWORKS****(Common to CSE, IT)****Time: 3 hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A**(25 Marks)**

- 1.a) What are the advantages of layered protocol suites? [2]
- b) Mention the comparison of fiber optics and copper wire. [3]
- c) Define Gigabit Ethernet. [2]
- d) Write about P-Persistent CSMA. [3]
- e) Write short note on design issues of network layer. [2]
- f) What are the approaches to apply on the congestion control? [3]
- g) What is meant by Internetwork routing? [2]
- h) Mention the differences between IPv4 and IPv6. [3]
- i) Write short notes on TELNET. [2]
- j) Write about client-server model in brief. [3]

PART - B**(50 Marks)**

2. Explain OSI reference models, and compare with TCP/IP reference model. [10]
- OR**
- 3.a) Explain sliding window protocol with an illustration.
 - b) Explain the Hamming Code with an example. [4+6]
4. Explain the following:
 - a) Ethernet MAC sub layer protocol.
 - b) ALOHA protocol. [5+5]
- OR**
5. Explain about the following devices:
 - a) Hub
 - b) Switches
 - c) Router
 - d) Gateway. [10]
6. Write in detail about Link-state routing algorithm and mention its advantages and disadvantages. [10]
- OR**
7. What are the principles of congestion control? Explain it with virtual circuit subnets. [10]

8. Describe about flow Control and buffering in transport layer. [10]

OR

9. Write in detail about the Internet control message protocol. [10]

10. How does the Remote procedure call works? Explain it. [10]

OR

11. Write short notes on the following:

a) HTTP

b) DNS.

[5+5]

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